

Tan Dai Ngo

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EDUCATION

University of Victoria | MEng, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Optimization for Machine Learning, Systems for Massive Datasets, Data Mining. **GPA:** 8.78 / 9.00.

University of Chicago | MS, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Distributed Systems, Databases, Algorithms, Applied Software Engineering, Networks.

University of Washington | BS, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Applied Mathematics, Informatics. **GPA:** 3.57 / 4.00. **Certificate:** Quantitative Managerial Economics.

EXPERIENCE

Software Developer | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Built and maintained REST endpoints across 5+ Roaming Business System microservices in Java and Spring Boot.
- Cut Cassandra query response times below a 3 second budget across multiple data sources.
- Traced 20+ cross service defects to their root cause and shipped fixes via Jenkins across 50+ releases.

Math and Computer Science Teacher | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms, object oriented design and software engineering fundamentals to 50+ students.
- Explained technical material to students with no prior background, adjusting the pitch in real time.

PROJECTS

Multi-Stage IoT Intrusion Detection | Python, XGBoost, PyTorch *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six classifiers on 206,000 flows with preprocessing refit on training rows, reordering the ranking.
- Calibrated binary XGBoost to a 1% FPR budget, reaching 99.5% recall, 0.94% FPR and 0.989 PR-AUC.
- Designed a four condition holdout that exposed session leakage as binary PR-AUC fell from 0.981 to 0.125.

Ocean Data ML Platform with RAG | Python, Spark, XGBoost, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Ingested 10.8 million records into Bronze, then Spark Silver processing into a 3.69M row Gold table.
- Containerized FastAPI inference and Qdrant retrieval with Docker Compose, at hit@k 0.90 over ten queries.
- Tuned XGBoost with Optuna against a persistence baseline, cutting air temperature RMSE 48% at 12 hours.

Fuel Blending Prediction Service | Python, CatBoost, FastAPI, AWS *Shell.ai Hackathon 2025 | Jul 2025*

- Engineered weighted aggregates, spread and fraction entropy across 10 properties of five component blends.
- Selected multi output CatBoost on five fold out of fold MAPE of 0.64 against XGBoost at 1.29.
- Packaged the fitted scaler and training medians with the model so serving matched training exactly.

Automated Crypto Trading Bot | Python, Coinbase API, AWS, pandas *Personal Project | Feb - Jul 2025*

- Implemented a mean reversion engine that buys 1% dips and sells on 2% rises over rolling 60 minute windows.
- Added a trend strategy that buys when the EMA over 8 periods tops the 20 and price clears the 200 EMA.
- Backtested both strategies on four months of Coinbase data, then deployed to EC2 with CSV risk controls.

TECHNICAL SKILLS

Languages & Programming: Python, Java, SQL, Go, JavaScript, Cypher, MATLAB, object oriented programming

Machine Learning & Evaluation: PyTorch, XGBoost, CatBoost, scikit-learn, pandas, NumPy, Optuna, MLflow

Data, Retrieval & Messaging: Spark, PySpark, ETL, Qdrant, RAG, embeddings, vector search, Cassandra, Kafka

Backend, Cloud, CI/CD & Testing: Spring Boot, FastAPI, REST, Docker, AWS (EC2, S3, Lambda), Jenkins, pytest

Credentials: Google Advanced Data Analytics (2025); Shell.ai Hackathon 2025; IMC 64 Bids, top 250 of 1,500